Spatial rescue in biofilms: a double-edged sword

May 17, 2026

Abstract

A biofilm under antibiotic treatment has a clear spatial asymmetry: cells on the surface are exposed to the drug, while cells in the deeper interior are shielded from it. We ask whether that shielded interior makes the population more likely to evolve drug resistance and survive, a process known as evolutionary rescue. Comparing a spatially structured biofilm model against a well-mixed population of the same initial size, and sweeping the antibiotic dose along with the metabolic activity of the interior, we find that the biofilm is a double-edged sword; it rescues less often than the well-mixed reference at low dose, but more often at high dose, and only when the interior is metabolically active. We explain this pattern as a consequence of two opposing effects of the spatial structure: a supply mechanism that generates resistant mutants in the interior yet is subject to drift since it is not exposed to the drug, and a delivery mechanism that transports those mutants to the surface where they can grow and establish but is tied to antibiotic dose via the velocity at which the biofilm deteriorates. Our results suggest that the spatial structure of biofilms can both hinder and facilitate evolutionary rescue, and that the balance between these effects depends on the antibiotic dose and the metabolic activity of the interior.

1 Introduction

Antibiotic resistance is an escalating clinical problem, and biofilms (dense, surface-attached bacterial communities) carry a disproportionate share of it. Biofilms are responsible for a large fraction of chronic infections and are notoriously hard to clear with antibiotics; standard regimens that work in well-mixed culture often fail when the target population is organized as a biofilm. This has driven interest in behavior-modifying interventions, in particular compounds that induce bacterial dispersal away from biofilms toward a planktonic, or more well-mixed state. Such interventions do not kill cells; they simply change how cells are arranged in space. The hope is that disrupting biofilm spatial structure reduces the population's capacity to evolve resistance under antibiotic pressure, essentially disarming the bacteria such that they will be more easily cleared by traditional antibiotic treatments and the host immune system. Knowing whether that hope is justified requires understanding what the spatial structure of a biofilm actually does to a population's evolutionary dynamics in the first place. Does biofilm formation help the population evolve resistance, hurt it, or some combination of both?

Biofilms have a clear spatial asymmetry: cells on the outer surface are exposed to the drug, while cells in the deeper interior are shielded from it. The shielded interior is a spatial refuge, a region of the population where the antibiotic selection pressure does not reach. We ask whether that refuge makes the population more likely to undergo *evolutionary rescue*, the outcome in which a declining population avoids extinction because a resistant mutant arises and takes over in time. Concretely: does a biofilm with a shielded interior rescue more often than a well-mixed population of the same initial size?

The naive prediction is that the refuge provided by biofilm structures should help facilitate evolutionary rescue. The biofilm creates a gradient of antibiotic concentration, with the shielded interior experiencing lower levels of the drug. This shielded interior lets the population persist for longer under treatment, and a longer overall population-level lifespan means more cell divisions, which means more opportunities for a resistant mutation to arise. More mutations should mean more rescue. But this prediction has at least two distinct ways it could be wrong. First, a mutation that arises in the shielded interior does nothing on its own: there is no selection pressure inside the refuge to make a resistant cell take over, so a core mutation matters only if it eventually reaches the drug-exposed surface in time to grow under selection. The refuge might generate mutations the population can never actually use. Second, the very mechanism that maintains the refuge, namely the replacement of dying surface cells by cells flowing outward from the interior, could act against a new resistant cell trying to establish on the surface by ensuring it is surrounded by a continuously refreshed wild-type majority. The same spatial structure that extends the population's lifetime may also make each individual mutation that arises less useful. Whether the biofilm is a net help or a net hindrance therefore depends on how these effects balance.

We measure that balance directly by simulating a minimum-viable biofilm (a one-dimensional chain of cells with a drug-exposed edge and a shielded core) under antibiotic pressure, and compare its rescue probability against an otherwise identical well-mixed population. By varying the antibiotic dose and the metabolic activity of the interior independently, we ask whether the biofilm's spatial structure is uniformly helpful, uniformly harmful, or, as the results below will show, contingent on both the selection pressure and on what is happening inside the refuge.

2 The model

We model the biofilm as a one-dimensional chain of cells. Real biofilms are three-dimensional, but the central asymmetry we are after (drug-exposed surface versus shielded interior) does not need three dimensions to capture. One dimension keeps the dynamics tractable while preserving the feature that matters: cells at different positions face different antibiotic concentrations.

2.1 The chain and rates

The state of the population at any time t is an ordered list of cells indexed $0, 1, \ldots, N(t) - 1$. Index 0 is the deep interior; index $N(t) - 1$ is the outer surface. Each cell has a genotype, either wild-type (WT) or resistant (R), and a position determined by its index. As per convention with evolutionary rescue, the population starts at $t = 0$ with $N(0) = N_0$ WT cells and no resistant cells. The population then evolves under antibiotic pressure, which begins at time $t = 0$, with births and deaths changing the chain's length and composition over time.

Antibiotic exposure is a step function in position. At any given population size $N(t)$, the rightmost l cells of the chain form the edge, the region where antibiotic concentration is lethal to WT. In this model, we assume l to be a fixed integer parameter describing the width of the edge, or the thickness of the drug-exposed layer. All cells with index $< N(t) - l$ form the core, the shielded interior. The boundary between edge and core sits at position

$$b(t) \;=\; \max\bigl(0,\, N(t) - l\bigr), \tag{1}$$

so as the population shrinks, the boundary retreats from the surface toward the interior. When $N(t) \le l$ the boundary collapses to $b = 0$, and the entire population is edge; no core remains. Additionally, the well-mixed equivalent of this model is the special case where $l \ge N_0$, so the edge is as wide as the entire population and there is no core at all. The boundary is always at $b = 0$ in that case, and every cell is drug-exposed from the start. The flat step-function profile is a stylization of the real, smooth antibiotic gradient through a biofilm. Treating the boundary as sharp is the minimum-viable abstraction that captures the asymmetry; so it is reasonable to assume that a smooth gradient would change quantitative numbers but not the qualitative dynamics we are after.

Each cell has a birth rate and a death rate determined by its compartment and its genotype, summarized in Table 1.

	Core ($i < b$)		Edge ($i \ge b$)	
	birth	death	birth	death
Wild-type (WT)	b_c	d_c	b_e	d_e
Resistant (R)	b_c	d_c	b_R	d_R

Table 1: Per-cell birth and death rates by compartment and genotype, where i is the cell index in the chain, and $b(t)$ is the boundary position between core and edge. The eight compartment-genotype-event combinations that drive the Gillespie dynamics are described in §2.2.

The edge WT birth rate $b_e = 1$ sets the time unit, where time is tracked in generations as the inverse of the birth rate $\frac{1}{b_e} = 1$. The edge WT death rate d_e is the antibiotic dose, swept over values $d_e > b_e$ so WT cells in the edge are always net-dying under treatment. Resistant cells in the edge have $b_R = 1$ and $d_R = 0.6$, giving them a positive net growth rate $s_R = b_R - d_R = 0.4$ once they reach the drug-exposed surface; an R lineage that establishes in the edge grows deterministically to dominance. Notably, however, there are different classes of antibiotic, and some drugs are bacteriostatic rather than bactericidal. For a bacteriostatic drug, the edge WT death rate would be $d_e = b_e = 1$, so WT cells in the edge would be net-neutral rather than net-dying. We do not study that case here, but it is straightforward and possibly interesting to implement by setting $d_e = 1$. [1] Additionally, the model allows multiple R lineages to arise and compete, but the assumptions are such that rescue is defined whenever there are at least $r_{\text{thr}} = 10$ R cells in the edge, so the first lineage or combination of lineages to reach that threshold is what determines rescue. As such, we do not examine the dynamics of clonal interference among R lineages, which would be an interesting topic for future work but is not the focus here.

Cells in the core, regardless of genotype, have balanced rates $b_c = d_c$. The core has no net growth on its own, and WT and R are treated identically there because the shielded interior carries no selection. The default core rate is $b_c = 0.2$, a slow but nonzero pace; the dormant-core limit is $b_c = 0$, where the interior produces no births and no deaths at all. Comparing these two limits is one of the central experiments below.[2]

[1] We would expect that if WT cells were stalled in growth but not dying, rescue dynamics would be different.

[2] To prevent unbounded post-rescue growth, R births in both compartments are multiplied by a Wilson-type

2.2 Dynamics

Events fire one at a time, sampled as a continuous-time Markov chain via the Gillespie algorithm. At each step we compute the total event rate R_{tot} across all eight compartment-genotype-event combinations in Table 1, advance simulated time by an exponential draw with mean $1/R_{\mathrm{tot}}$, and pick which event fires from a categorical distribution with weights proportional to each event's rate.

A birth event picks a parent uniformly at random within its compartment-genotype subgroup. The daughter is inserted adjacent to the parent (at the parent's position or one position to the right, each with probability $1/2$), and the chain grows by one. A death event picks a victim uniformly at random within its subgroup and removes it; the chain contracts by one.

The boundary $b(t)$ is recomputed after every event according to Eq. (1). When the population grows by one and the boundary moves outward, the cell that now sits at the new boundary position has crossed from edge into core; it is reclassified as a core cell. When the population shrinks and the boundary moves inward, the cell that now sits one position outside the new boundary has crossed from core into edge; it is reclassified as an edge cell. These reclassifications are not separate dynamical events: they are bookkeeping consequences of recomputing the boundary against a chain that has just changed length by one.

This bookkeeping is the *conveyor belt* mechanism the rest of the writeup will refer to. As long as the core has cells in it, every edge death pulls an interior cell out to the surface, and every edge birth pushes an outer cell back into the interior. The result is that the edge holds exactly l cells while the core has anything left to supply; the population is shrinking, but the surface layer is being continuously replaced from below. Figure 1(a, b) illustrates one such reclassification on a small didactic chain: a cell in the same physical position is in the core before an edge death and in the edge after, because the boundary has moved over it.

The conveyor belt produces two qualitatively different regimes of decline. While the core still has cells in it ($N(t) > l$), each edge death is replaced immediately by a core cell, so the edge stays at exactly l cells. Only the l edge cells contribute to net loss, at rate $s_e \cdot l$ per unit time, and the total population falls linearly, where $s_e = d_e - b_e$. The duration of this buffered decline is

$$T_1 \;=\; \frac{N_0 - l}{s_e \cdot l}, \tag{2}$$

which is long at low dose (slow edge turnover empties the core gradually) and brief at high dose (fast edge turnover empties the core quickly). Once the core is exhausted ($N(t) \leq l$; Figure 1(c)), the edge is the entire remaining population, and there is no reservoir to draw on. The remaining $\leq l$ cells decay exponentially at rate s_e, just as a well-mixed population would. The transition between these two regimes is sharp on the simulation timescale: it happens at the moment the last core cell is converted to edge. After that point, the dynamics on the chain are indistinguishable from those of a one-compartment well-mixed model with $N = l$.

2.3 Mutations, rescue, and initial condition

Mutations are replication-linked: at each WT birth event, with probability μ, the daughter is an R cell instead of WT; otherwise the daughter inherits the parent's genotype. Mutations are

density factor $\max(0, 1 - N/K)$, with $K = N_0$ throughout. WT births stay density-independent. The factor is approximately 1 during the wild-type decline that we study, so it does not affect the dynamics of interest; it matters only once R has taken over.

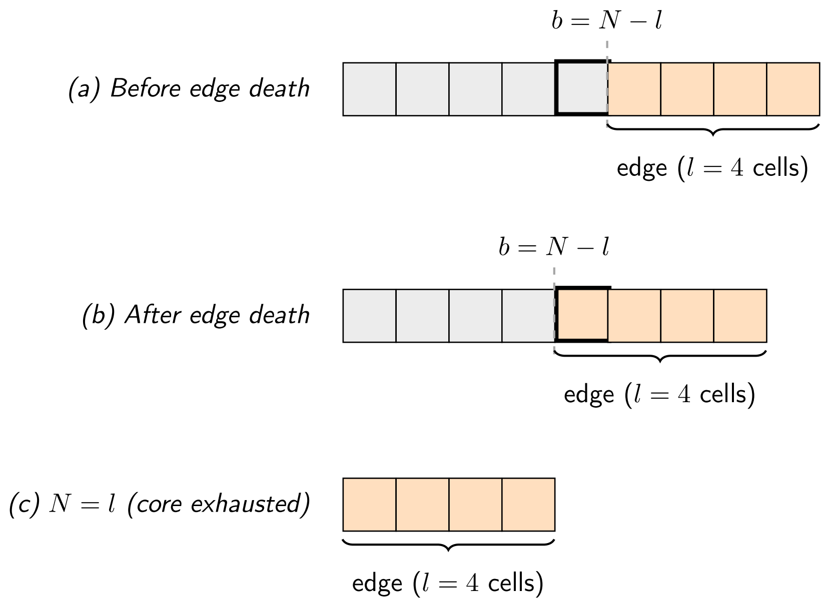

Figure 1: The conveyor belt as a consequence of recomputing the boundary, illustrated on a small didactic chain with $N = 9$ initial cells and $l = 4$. The rightmost l cells (orange) form the edge; the rest (grey) form the core. (a) Before the event, the cell drawn with a heavy outline sits in the core, immediately inside the boundary $b = N - l$. (b) After an outer edge cell has died, N has dropped to 8 and the recomputed boundary has moved one position inward; the tracked cell, in the same physical position, is now part of the edge. No cell has been shuffled; the bracket has shifted over it. (c) After enough cumulative edge deaths, N reaches l: the boundary collapses to $b = 0$, the entire chain is edge, and the dynamics enter Phase 2. An edge birth is the symmetric operation, with the boundary moving outward by one position and the innermost edge cell reclassified as core.

irreversible (no back-mutation from R to WT), so once a lineage is R, all of its descendants are R. Because mutations ride on births, they appear at specific spatial positions on the chain (the position of the dividing parent); a mutation born deep in the core enters the world at the core's spatial location and must be physically transported to the edge to come under selection.

Every R lineage that ever appears is tagged with a unique identifier when its founder is born, and that identifier propagates to all descendants. This lets us follow each mutation individually: where it was born (core or edge), when it appeared, how its descendant population grew, whether it ever reached the edge, and whether it contributed to rescue. The per-lineage records produced by this tracking are the basis for most of the analyses in later sections.

A run is declared $\mathrm{rescued}$ when the edge contains at least $r_{\mathrm{thr}} = 10$ resistant cells simultaneously. The choice of $r_{\mathrm{thr}} = 10$ is high enough to be well past the establishment bottleneck (the regime where a small R lineage might still go extinct by chance) but small enough that rescue is declared promptly after takeover begins. Once an R lineage is past establishment in the edge, where $s_R > 0$, it grows deterministically to dominance. Every simulation terminates naturally on one of two events: rescue ($\geq r_{\mathrm{thr}}$ R cells in the edge) or extinction ($N = 0$). There are no artificial time or generation caps. Both endpoints are guaranteed by the dynamics: either a successful resistant lineage establishes in time, or the wild-type population is driven to zero. The model cannot stall in between.

Every simulation starts at $t = 0$ with $N = N_0$ wild-type cells and no resistant cells. Before $t = 0$ the population is at quasi-steady state, with balanced birth and death rates in both compartments ($b_e = d_e = 1$ pre-treatment in the edge, $b_c = d_c$ in the core), so the biofilm neither grows nor shrinks. At $t = 0$ the antibiotic switches on: the edge WT death rate jumps from $b_e = 1$ to its treatment value $d_e > 1$, and the population begins to decline. What we simulate is therefore the response of an already-mature biofilm to a step-function antibiotic challenge. Real treatments are gradual; this abstraction strips out the loading dynamics so that the rescue dynamics, the only thing we are studying, are not entangled with growth-phase or pharmacokinetic transients.

2.4 The well-mixed reference

To say whether the biofilm helps or hurts, we need a baseline. Our baseline is a well-mixed population of the same initial size N_0, governed by the edge birth and death rates alone: every cell is drug-exposed, none are shielded. This is the standard setting for evolutionary rescue (Wilson et al., 2017), and it is the natural null model, the same biofilm with all spatial structure removed. We simulate this fully well-mixed model (essentially a one-compartment model and faithful recreation of Wilson's evolutionary rescue model). Operationally, however, the true well-mixed reference to our biofilm model is the chain model with $l \geq N_0$: when the edge is as wide as the entire population, no cell is ever in the core, the conveyor belt never engages, and the dynamics reduce to a one-compartment birth-death process on scalar counts. We also maintain an independent scalar-count implementation as a cross-check; the two well-mixed models agree statistically within sampling error, which serves as an internal consistency test on the chain model.

2.5 Parameters and simulation details

Unless stated otherwise, all simulations use the parameters in Table 2. Time is measured in edge generations ($b_e = 1$). The dose sweep covers $d_e \in \{1.05, 1.1, 1.2, 1.4, 1.6, 1.9, 2.3, 2.8\}$. The core-activity sweep covers $b_c \in \{0, 0.05, 0.1, 0.2, 0.4, 0.8\}$. The edge-width sweep covers $l \in \{25, 50, 100, 200, 400\}$. Every parameter combination is run with $5{,}000$ independent seeds; the well-mixed reference is run at the same dose points with the same seed count.

Parameter	Value	Meaning
N_0	1000	Initial population size
l	100 (swept)	Fixed value for edge width (# cells exposed to drug)
b_e	1.0	WT birth rate in edge (sets time unit)
d_e	dose (swept)	WT death rate in edge; super-MIC is $d_e > 1$
b_R	1.0	R birth rate in edge
d_R	0.6	R death rate in edge ($s_R = 0.4$)
$b_c = d_c$	0.2 (swept)	Core birth & death rate ($0 =$ dormant)
μ	10^{-4}	Mutation probability per birth
r_{thr}	10	Resistant edge cells required to declare rescue
K	N_0	Density cap for R births (see footnote, §2.1)

Table 2: Default model parameters. The swept variables are dose (d_e), core activity (b_c), and edge width (l).

3 The double-edged sword

Section 1 closed with a naive prediction: the biofilm's refuge should make rescue more likely than in the well-mixed reference. We also flagged two distinct ways that prediction could be wrong. The geometric refuge could be useless on its own, with the extra lifetime never translating into extra rescues. Or the refuge could be helpful only under specific conditions, perhaps only at high dose or only when the interior is metabolically active, so that an "average" effect across regimes would smear out a sharp contingency. We want to distinguish these failure modes from each other, and from the naive prediction, in a single experiment.

Two knobs, varied independently, do the job. We vary the antibiotic dose between a value just above the minimum inhibitory concentration ($d_e = 1.05$) and a value well above it ($d_e = 2.8$); and we vary the metabolic activity of the interior between $b_c = 0$ (dormant, with no births or deaths inside the refuge) and $b_c = 0.8$ (active, with the interior dividing at a substantial fraction of the edge rate). Each knob isolates a different question. The dormant-core column tests whether the geometric refuge by itself matters: with no births inside, the only thing the biofilm has over the well-mixed reference is extra lifetime, so any increase in rescue probability here would have to come from time alone. The active-core column tests whether what happens inside the refuge matters: the interior is now dividing, adding cell divisions on top of the lifetime extension. Running both at low and high dose then asks whether either effect depends on the selection pressure that is forcing rescue in the first place. In Figure 2 we ran $5{,}000$ independent simulations of the biofilm and $5{,}000$ of the well-mixed reference per parameter combination.

Figure 2 reduces to two observations. First, Panel A shows probability of rescue for both the biofilm and the well-mixed reference at low and high dose, assuming a dormant core ($b_c = 0$), meaning that the interior is not actively dividing or dying (i.e., no turnover of cells in the core). The biofilm rescues less often than the well-mixed reference at both doses, with no hint of a crossover. The same spatial structure that extends the population's lifetime in a biofilm reduces rather than raises the rescue probability, and it does so at both low and high dose. Thus, extra lifetime alone does not convert into more rescue at any dose we tested.

Second, Panel B shows the same comparison but with an active core ($b_c = 0.8$), meaning that the interior is metabolically active and cells are dividing and dying at a substantial fraction of the edge rate. The dormant-core pattern from Panel A still holds at low dose: the biofilm rescues less often than the well-mixed reference. But at high dose, the pattern flips: the biofilm rescues roughly twice as often as the well-mixed reference. The same spatial structure that reduces the rescue probability at low dose raises it at high dose, allowing biofilms to rescue more often than the well-mixed reference at high dose.

Both observations, especially the second, contradict the naive intuition. The dormant column (Panel A) already shows that extra lifetime alone does not produce more rescue, as the biofilm rescues less often than the well-mixed reference at both low and high dose. The active column (Panel B) then shows that the interior's metabolic activity is required for the biofilm to rescue more often than the well-mixed reference, and this only happens at high dose (i.e., high d_e), where the antibiotic is most aggressively driving the population down. The same mounting selection pressure that should make rescue harder for everyone is precisely where the biofilm rescues more often than the well-mixed reference. We call this phenomenon the *double-edged sword*: the same spatial structure provided by a biofilm reduces the rescue probability at low dose but raises it at high dose, and two conditions, an active interior and a high antibiotic dose, have to coincide before the biofilm rescues more often than the well-mixed reference at all.

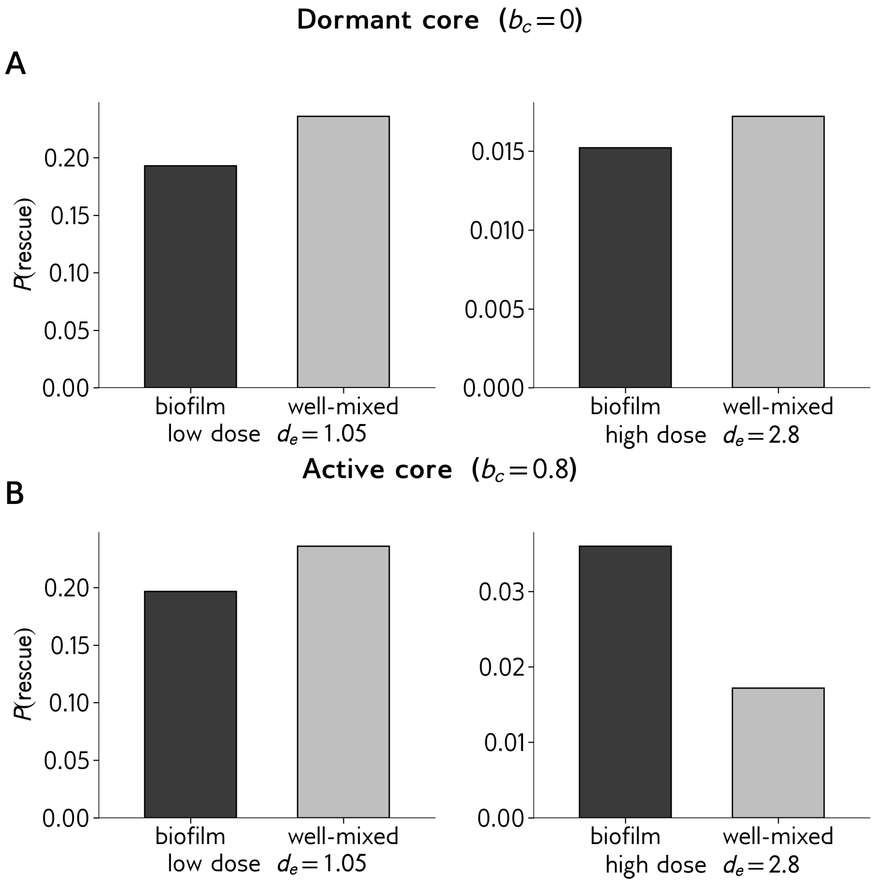

Figure 2: Rescue probability for the biofilm (dark) versus the well-mixed reference (light), at low dose ($d_e = 1.05$, left columns) and high dose ($d_e = 2.8$, right columns). Panel A: dormant core ($b_c = 0$). Panel B: active core ($b_c = 0.8$). Each subplot uses its own y-axis scale. *Parameters*: $N_0 = 1000$, $l = 100$, $\mu = 10^{-4}$; each bar averages over $5{,}000$ independent simulation runs.

The most pressing follow-up is what the active core is contributing at high dose that it fails to contribute at low dose, and why the dormant biofilm gets no benefit from its extra lifetime in the first place. Both questions concern where rescue actually comes from inside the biofilm. The next section opens up rescue itself by separating it into the two things any population needs in order to rescue: a resistance mutation has to arise (supply), and once it does, it has to grow to dominance in the drug-exposed edge, where antibiotic selection makes resistance an advantage (establishment). Tracking these two ingredients across the same conditions as Figure 2 is what lets us see where the biofilm and the well-mixed reference actually diverge.

4 Supply and establishment

Section 3 ended by separating rescue into two requirements: a resistance mutation has to arise (supply) before extinction, and once it does, it has to grow to dominance in the drug-exposed edge, where antibiotic selection makes resistance an advantage (establishment). To understand where the dose-dependent flip in Figure 2 comes from, we measure each requirement separately. The comparison between biofilm and well-mixed then splits into two pieces: how many mutations each population produces, and what fraction of those mutations establish in the edge.

Before splitting rescue into its two pieces, we first check that the bar-plot flip is a continuous feature of the dose response, not an artifact of sampling at two specific doses. Figure 3 Panel A shows the three populations across the full dose sweep. The active-core biofilm here is drawn at our default $b_c = 0.2$, which sets the core's birth and death rate to one-fifth of the edge birth rate $b_e = 1$; this is considerably less active than the $b_c = 0.8$ used for Figure 2, where the core divides and dies at four-fifths of the edge rate. The active-core biofilm crosses the well-mixed reference smoothly near $d_e \approx 1.4$ and stays above it at every higher dose; the dormant-core biofilm ($b_c = 0$, no metabolism in the interior) sits below the well-mixed reference at every dose. Panel B plots the same data as differences from the well-mixed reference: the active biofilm's $\Delta P(\text{rescue})$ crosses zero near $d_e \approx 1.4$, and the dormant biofilm's $\Delta P(\text{rescue})$ stays below zero throughout the sweep. Both bar-plot patterns therefore persist across the full dose range, and the crossover is robust to the level of core activity: even at the more modest $b_c = 0.2$, an active interior is enough to flip the comparison at high dose.

With the crossover confirmed across the full dose range, we still need to explain why the biofilm rescues differently from the well-mixed reference at all, and why the difference flips sign with dose. Following the framework set up at the start of this section, any difference in rescue probability between the two populations must come from a difference in how many R mutations each population produces, in what fraction of those mutations actually establish in the drug-exposed edge, or in both. Localizing the difference to one of these two factors (or finding it in both) is what lets us move past describing the flip and start explaining it. Figure 4 measures both differences side by side: how mutation production compares between biofilm and well-mixed (Panel A), and how the chance of an edge-born R mutation establishing compares (Panel B). Panel A reports the first comparison as the difference in mutation count $\Delta M = M_{\text{BF}} - M_{\text{WM}}$ between the biofilm and the well-mixed reference, where M is the mean number of R mutations a population produces per simulation run. Two observations. First, the dormant-core biofilm sits essentially on the $y = 0$ reference line at every dose: the dormant biofilm and the well-mixed reference produce essentially the same number of mutations, even though the dormant biofilm lives longer. We return to why that extra lifetime does not translate into extra mutations in a later section. Second, the active-core biofilm sits above the $y = 0$ line at every dose: an actively dividing interior produces more mutations than the well-mixed reference at every dose. The number of extra mutations the active biofilm produces over the well-mixed reference shrinks with dose, from roughly 1.75 extra mutations per run at $d_e = 1.05$ down to ~ 0.05 at $d_e = 2.8$, because the absolute number of mutations any population produces shrinks at higher dose (shorter lifetimes give fewer cell divisions during which a mutation can occur). In ratio terms, however, the active biofilm produces about 1.8 to $1.9\times$ as many mutations as the well-mixed reference at every dose. Notably, the number of extra mutations the active biofilm produces does not track where the biofilm wins: the largest excess in mutation count is at low dose, precisely where the active biofilm rescues less often than the well-mixed reference (Figure 3 Panel B); the excess shrinks toward zero at high dose, where the biofilm actually rescues more often. Where the active biofilm has the most additional mutations available, it converts them into the fewest rescue events.

Figure 4 Panel B reports the second piece: the difference in per-mutation establishment between biofilm and well-mixed, $\Delta P(\text{est}) = P_{\text{est, BF}} - P_{\text{est, WM}}$, where $P(\text{est})$ is the fraction of

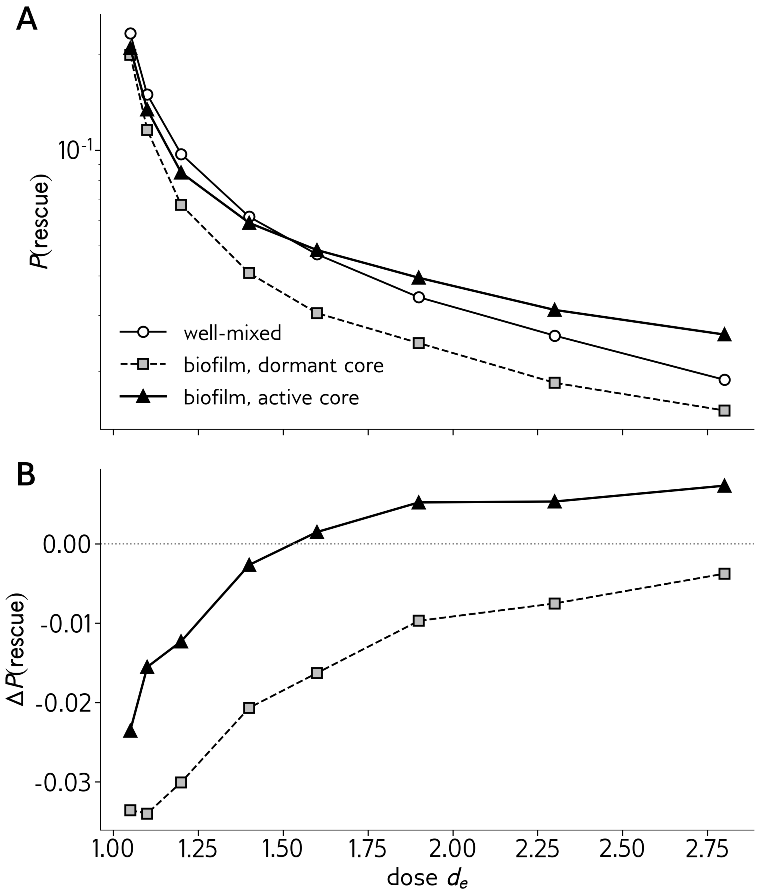

Figure 3: Probability of rescue versus dose for the three populations: well-mixed reference (open circles), biofilm with a dormant core ($b_c = 0$, open squares), and biofilm with an active core ($b_c = 0.2$, filled triangles). Panel A: absolute $P(\text{rescue})$, log y-axis. Panel B: difference relative to the well-mixed reference, $\Delta P(\text{rescue}) = P_{\text{BF}} - P_{\text{WM}}$, with the dotted horizontal line at zero marking the no-difference baseline. Each point averages $100{,}000$ independent simulations per parameter combination. *Parameters*: $N_0 = 1000$, $l = 100$, $\mu = 10^{-4}$.

edge-born R lineages that successfully establish rather than going extinct. Both biofilm curves sit below the $y = 0$ reference line at every dose: a given edge-born R lineage is consistently less likely to establish inside a biofilm than in the well-mixed reference, by roughly 0.05 to 0.10 in absolute probability across most of the dose range. This reduced establishment applies to both dormant and active biofilms and persists across the entire sweep. We return to why edge-born R lineages are less likely to establish inside a biofilm in a later section.

Putting Figure 4 together with Figure 3 reveals a tension. The active biofilm produces about $1.8\times$ as many mutations as the well-mixed reference at every dose, and every biofilm mutation establishes about $0.8\times$ as often. Multiplying these two ratios predicts that the active biofilm should rescue about $1.5\times$ as often as the well-mixed reference at every dose. But Figure 3 shows the actual rescue ratio is sharply dose-dependent: it climbs from below $1\times$ at $d_e = 1.05$ (the biofilm rescues less than well-mixed) to about $2\times$ at $d_e = 2.8$ (the biofilm rescues more). Neither the

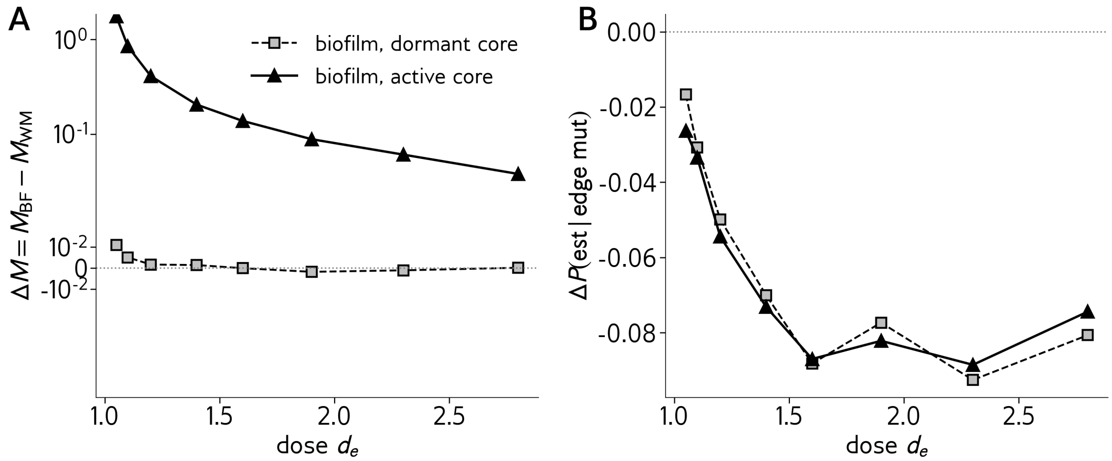

Figure 4: Decomposing the biofilm-to-well-mixed comparison into the two ingredients rescue requires. Panel A: difference in mutation supply $\Delta M = M_{\mathrm{BF}} - M_{\mathrm{WM}}$, where M is the mean number of R mutations a population produces over a simulation run. Symlog y-axis, with the dotted line at $y = 0$ marking equal supply. Panel B: difference in per-mutation establishment fraction for edge-born mutations, $\Delta P(\mathrm{est}) = P_{\mathrm{est,\,BF}} - P_{\mathrm{est,\,WM}}$, where $P(\mathrm{est})$ is the fraction of edge-born R lineages that successfully establish rather than going extinct. Linear y-axis, with the dotted line at $y = 0$ marking equal establishment. Both panels show two biofilm conditions across the dose sweep: dormant core ($b_c = 0$, open squares) and active core ($b_c = 0.2$, filled triangles). Each point averages $100{,}000$ independent simulations per parameter combination. *Parameters*: $N_0 = 1000$, $l = 100$, $\mu = 10^{-4}$. The decomposition needs more replicates than the rescue curves in Figure 3 because $P(\mathrm{est})$ is computed as a fraction of the already-rare edge-born mutation events, which become rarer at higher dose; even at $100{,}000$ runs per parameter combination, some residual noise remains visible in Panel B at $d_e \geq 1.5$.

supply ratio nor the establishment ratio depends on dose, yet the rescue ratio plainly does. The dose-dependent flip we set up in Section 3, and that this section's two-panel decomposition was supposed to localize, must therefore live somewhere we have not yet measured.

To find that missing piece we need to look at where the active biofilm's extra mutations are actually born. They come overwhelmingly from the core, where there is no antibiotic selection: a core-born resistant cell has no growth advantage over a wild-type core cell. A core-born mutation can only contribute to rescue if it eventually reaches the drug-exposed edge before going extinct, and Panel B's per-mutation establishment fraction does not account for this passage from core to edge: it reports establishment for edge-born mutations only. The fraction of core-born mutations that reach the edge depends on how fast the retreating boundary catches up to them, and the speed of that retreat is itself dose-dependent. The next section measures this fraction directly. It is the piece our decomposition has not yet captured, and once we have it, the explanation of the rescue flip set up in Section 3 is complete.

5 Reaching the edge

Section 4 ended on a tension. The two factors we measured (how many R mutations each population produces, and what fraction of those mutations succeed at establishing in the edge) both differ between biofilm and well-mixed in roughly dose-independent ways: the active biofilm produces about $1.8\times$ as many mutations as the well-mixed reference at every dose, and each biofilm mutation establishes about $0.8\times$ as often at every dose. Multiplied together, these two ratios predict a constant biofilm-to-well-mixed rescue ratio. The actual rescue ratio is not constant. It climbs from below $1\times$ at $d_e = 1.05$ to about $2\times$ at $d_e = 2.8$. Some dose-dependent factor that those two measurements did not capture must therefore be carrying that flip. The candidate is the one piece of the rescue calculation that the two panels of Figure 4 do not cover: whether a core-born R mutation ever reaches the drug-exposed edge before going extinct. The supply panel counts every R mutation a population produces, but a core-born mutation contributes to rescue only if it arrives in the edge, where antibiotic selection makes resistance an advantage. The establishment panel measures what happens to mutations once they are in the edge, but not whether they got there. Whether a given core-born R lineage actually reaches the edge is therefore a separate per-lineage outcome from either of the two pieces Section 4 measured, and we measure it directly across the same dose sweep.

Figure 5 Panel A reports this fraction directly. Two curves are shown: the default active-core biofilm ($b_c = 0.2$, open squares) and the more active core used in Section 3's bar plot ($b_c = 0.8$, filled triangles). Both curves rise sharply with dose. At $d_e = 1.05$, only about 5 to 10 percent of core-born R lineages ever reach the edge; at $d_e = 2.8$, that fraction climbs to between 40 and 80 percent depending on core activity. At every dose the less active core actually has a higher fraction of core lineages reaching the edge than the more active core, an inversion we return to in the mechanism discussion below. The main point is the dose-dependent shape: in both cases the fraction rises sharply between low and high dose, matching the rescue flip in Figure 3.

The shape of Panel A follows from a race between two processes inside the core. A core-born R lineage starts as a single cell with balanced birth and death rates ($b_c = d_c$), so it does not grow on average; instead it fluctuates in size and, given enough time and turnover of cells in the core, is likely to go extinct by drift. The only way the lineage can matter for rescue is if the retreating boundary reaches its position before drift kills it. The boundary moves inward at speed $s_e \cdot l$ per unit time, where $s_e = d_e - b_e$ is the net edge death rate. At high dose, s_e is large; the boundary moves inward quickly and overtakes core lineages before drift has had time to eliminate them. At low dose, s_e is small; the boundary creeps inward very slowly, and most core lineages drift to extinction before they are ever exposed to the edge. The dose-dependence visible in Panel A is just this race playing out across the dose sweep, where the speed of population decline, matched with the relative size of the edge l, sets the time available for a core lineage to survive before the boundary catches it.

The same race explains a feature of Panel A that runs against naive expectation. At every dose, the less active core ($b_c = 0.2$) has a higher fraction of core lineages reaching the edge than the more active core ($b_c = 0.8$), even though the more active core produces roughly four times more core mutations per unit time. One might expect that more mutations would mean more chances for at least one to survive the race to the edge. But boundary speed and core drift are decoupled. Boundary retreat is set entirely by edge dynamics, so the time a lineage waits for the boundary to reach it is the same at $b_c = 0.2$ as at $b_c = 0.8$. The strength of drift in the core, however, depends directly on b_c: a more active core means more births and deaths per unit time

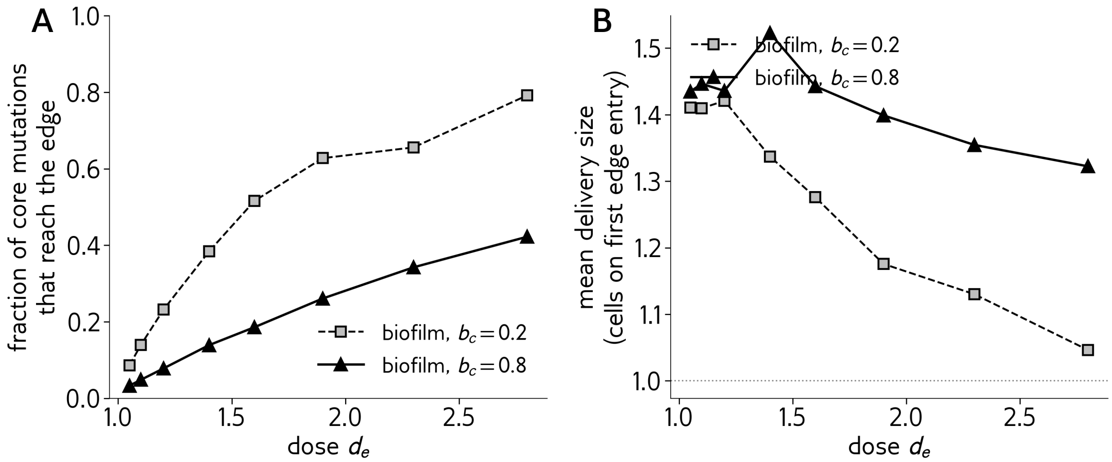

Figure 5: Per-lineage outcome of core-born R mutations. Panel A: fraction of core-born R lineages that ever reach the drug-exposed edge before going extinct, vs dose. Panel B: mean number of cells in a lineage when it first occupies an edge position, vs dose. Both panels show the default active core ($b_c = 0.2$, open squares) and the more active core ($b_c = 0.8$, filled triangles) used for Figure 2. Each point averages $100{,}000$ runs for $b_c = 0.2$ and $5{,}000$ runs for $b_c = 0.8$ per parameter combination. *Parameters*: $N_0 = 1000$, $l = 100$, $\mu = 10^{-4}$.

per cell, which means faster fluctuations in lineage size and faster extinction by chance. With $b_c = 0.8$, lineages are killed in the core more quickly than at $b_c = 0.2$, so a smaller fraction survive long enough for the boundary to reach them. The more active core produces more mutations in absolute terms, but it also kills them off faster, and the fraction surviving long enough to reach the edge drops with b_c even as the absolute mutation count rises with b_c.

The dormant case is the limit of this mechanism at $b_c = 0$. With no births in the core, no R lineages are ever born there in the first place; the race has nothing to enter. The boundary retreats at the same speed $s_e \cdot l$ as in the active case, but with no core mutations to catch, no core-born lineages ever reach the edge at any dose. The dormant biofilm therefore has no rescue advantage at any dose, not because the boundary fails to retreat, but because the core never produces a mutation for the boundary to expose to the edge in the first place.

Panel B reports a secondary consequence of the same race. The mean size of a core-born R lineage when it first arrives in the edge is slightly larger at low dose than at high dose: about 1.4 cells per arrival at $d_e = 1.05$, dropping toward 1 cell at higher dose. A lineage that survives a long wait in the core (which happens at low dose, when the boundary moves slowly) has had more time to fluctuate upward in size by chance before being caught. A lineage caught quickly (at high dose, when the boundary moves fast) is more likely to still be a single cell on first arrival. The dose dependence in Panel B runs opposite to what the rescue flip needs. A lineage arriving in the edge as a cluster of two or three cells has more independent chances to escape early extinction than a single-cell arrival: each cell can carry the lineage forward even if its siblings die. Larger clusters are therefore advantageous for establishment, and Panel B shows that larger clusters happen at low dose. If Panel B were the dominant influence on the rescue flip, the biofilm would rescue more often at low dose than at high dose. But the rescue flip goes the other way: the biofilm rescues less often at low dose and more often at high dose. Panel B therefore cannot be carrying the dose-dependence of the rescue flip.

What dominates instead is Panel A, and the reason is in the magnitudes. Panel A spans an order of magnitude across the dose range (from a few percent reaching the edge to about half reaching). Panel B varies by less than fifty percent (from about 1.4 down to about 1.0 cells per arrival). At low dose, the modest cluster advantage of a 1.4-cell arrival cannot offset the fact that almost no core-born R lineages are arriving in the first place. The dose-dependence of the rescue flip therefore lives in the fraction reaching the edge at all, not in the size of those arrivals. Putting the three measurements together accounts for the rescue flip from Section 3. Rescue requires either an edge-born R mutation to establish (the path open to all populations) or a core-born R mutation to reach the edge and then establish (the path open only to biofilms with an active core). The first path is similar in biofilm and well-mixed: edge mutation counts overlap, and the per-mutation establishment fraction is only slightly reduced for the biofilm. The second path is what carries the dose-dependence.

At low dose, fewer than ten percent of core-born R lineages ever reach the edge. The active core produces more mutations overall, but those extra mutations stay trapped in the core where they cannot contribute to rescue. The biofilm therefore rescues less often than well-mixed at low dose, because the slight reduction in per-mutation establishment is no longer offset by anything from the core. At high dose, almost half of core-born R lineages reach the edge, so the active core's extra mutations now contribute a meaningful number of R lineages to the edge population, and the biofilm rescues more often than well-mixed. The crossover near $d_e \approx 1.4$ is where the dose-dependent reach-the-edge fraction balances the dose-independent reduction in per-mutation establishment.

The dormant case follows from the same accounting. With no core births, the second path is closed; only the first path exists, with the same edge mutation count as well-mixed and the same slightly reduced per-mutation establishment as the active biofilm. So the dormant biofilm rescues less often than well-mixed at every dose, exactly what Figure 3 shows. The flip we set up in Section 3 is the active biofilm's second path overtaking the first at high dose, and the dormant biofilm's lack of a second path at any dose. What remains to test is whether this account holds across the model parameters we have so far held fixed: the biofilm geometry l, the core activity b_c, and the mutation rate μ. The next section sweeps each in turn.

6 Robustness of the rescue flip

Sections 3 through 5 have built and tested the rescue flip at one specific parameter combination ($N_0 = 1000$, $l = 100$, $\mu = 10^{-4}$). A reader who has accepted the race mechanism in Section 5 would reasonably ask whether the rescue flip is a robust qualitative feature of the model, or an artifact of this one setting. We sweep the three parameters held fixed in those sections (edge width l, core activity b_c, and mutation rate μ) and ask two questions of each sweep: does the flip persist across the parameter range, and does the crossover dose shift in ways the race mechanism from Section 5 would predict?

Figure 6 sweeps the edge width l across $\{25, 50, 100, 200, 400\}$, holding all other parameters at their defaults. Panel A shows the dose curves for the biofilm at each l alongside the well-mixed reference; Panel B shows the corresponding $\Delta P(\text{rescue}) = P_{\text{BF}} - P_{\text{WM}}$. The flip persists across a sixteen-fold change in edge width, but its magnitude scales strongly with l. At smaller l the flip is sharper, with the biofilm's high-dose advantage growing as l shrinks: at $d_e = 2.8$, $\Delta P \approx 0.013$

for $l = 25$ but is essentially zero at $l = 400$, where the biofilm and well-mixed populations are indistinguishable across the entire dose range. The crossover dose itself shifts only modestly, from roughly $d_e \approx 1.5$ at $l = 200$ down to $d_e \approx 1.4$ at $l = 25$.

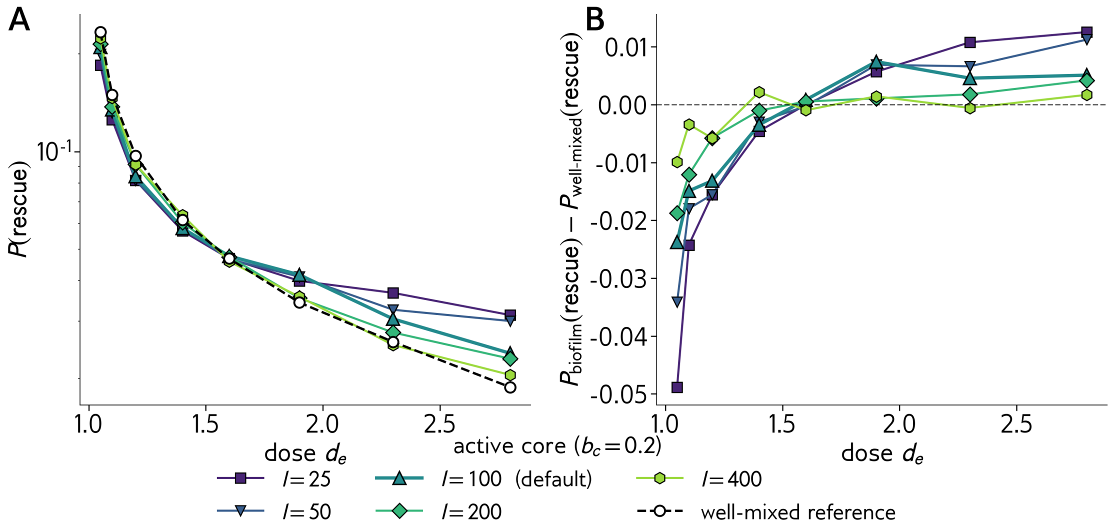

Figure 6: Sensitivity of the rescue flip to edge width l. Panel A: $P(\text{rescue})$ vs dose for the biofilm at five values of $l \in \{25, 50, 100, 200, 400\}$ and the well-mixed reference (open circles, dashed). Panel B: difference relative to the well-mixed reference, $\Delta P(\text{rescue}) = P_{\text{BF}} - P_{\text{WM}}$, with the dotted horizontal line at zero marking the no-difference baseline. Each point averages $20{,}000$ independent simulations per parameter combination. *Parameters*: $N_0 = 1000$, $b_c = 0.2$, $\mu = 10^{-4}$.

Both shifts can be read through the race mechanism from Section 5. Smaller l has two competing effects on the race. The boundary speed is $s_e \cdot l$, so smaller l slows boundary retreat in absolute cell terms, which by itself would reduce the fraction of core mutations that reach the edge. But smaller l also makes the core larger relative to the total population (it is $N_0 - l$ cells), which means more core births per unit time and a longer Phase 1 over which to produce them: the duration $T_1 = (N_0 - l)/(s_e \cdot l)$ increases as l decreases. The larger pool of core mutations more than compensates for the slower boundary retreat. The net effect tilts the race toward the biofilm at smaller l, which is what Panel B shows.

Figure 7 sweeps the core metabolic activity b_c across $\{0, 0.05, 0.1, 0.2, 0.4, 0.8\}$, holding all other parameters at their defaults. Panel A shows the dose curves for the biofilm at each b_c alongside the well-mixed reference; Panel B shows the corresponding $\Delta P(\text{rescue})$. At $b_c = 0$ (dormant core), the biofilm stays strictly below the well-mixed reference at every dose; there is no crossover. At every $b_c > 0$ in the sweep, the biofilm crosses the well-mixed reference at some dose and stays at or above it at higher doses. The crossover dose shifts predictably with b_c: from $d_e \approx 1.6$ at $b_c = 0.05$ down to $d_e \approx 1.2$ at $b_c = 0.8$. The high-dose advantage also grows with b_c: at $d_e = 2.8$, ΔP at $b_c = 0.8$ is about four times its value at $b_c = 0.05$.

The shifts can be read through the race mechanism from Section 5. The dormant case ($b_c = 0$) is the cleanest test: with no core births, no R lineages are ever produced in the core for the boundary to catch, and the biofilm gains no advantage at any dose. The $b_c > 0$ cases trade off two effects of increasing the core's metabolic rate. Higher b_c produces more core mutations per unit time, which

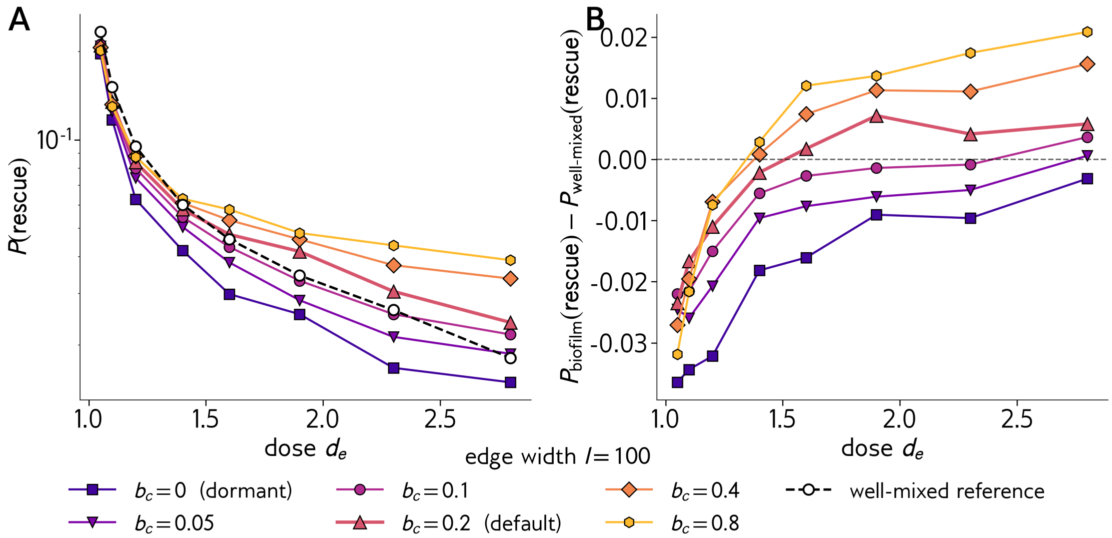

Figure 7: Sensitivity of the rescue flip to core metabolic activity b_c. Panel A: $P(\text{rescue})$ vs dose for the biofilm at six values of $b_c \in \{0, 0.05, 0.1, 0.2, 0.4, 0.8\}$ and the well-mixed reference (open circles, dashed). Panel B: difference relative to the well-mixed reference, $\Delta P(\text{rescue}) = P_{\text{BF}} - P_{\text{WM}}$, with the dotted horizontal line at zero marking the no-difference baseline. Each point averages $20{,}000$ independent simulations per parameter combination. *Parameters*: $N_0 = 1000$, $l = 100$, $\mu = 10^{-4}$.

feeds more lineages into the race. But higher b_c also accelerates drift in the core, so a smaller fraction of those lineages survive long enough for the boundary to reach them (Figure 5 showed this directly: at $b_c = 0.8$, fewer core lineages reach the edge as a fraction than at $b_c = 0.2$). The absolute mutation count rises faster than the fraction lost, so the total flow of core lineages arriving at the edge increases with b_c, and the biofilm's high-dose advantage grows accordingly.

Figure 8 sweeps the mutation rate μ across $\{10^{-5}, 10^{-4}, 10^{-3}\}$, holding all other parameters at their defaults. Mutation rate scales the absolute rescue probability strongly: at $\mu = 10^{-3}$ both populations rescue close to a substantial fraction of the time at low dose, while at $\mu = 10^{-5}$ rescue is rare for both at every dose. The qualitative flip persists at $\mu = 10^{-4}$ (our default, with crossover at $d_e \approx 1.4$) and at $\mu = 10^{-3}$ (crossover at $d_e \approx 1.5$, with the high-dose advantage about an order of magnitude larger than at the default μ). At $\mu = 10^{-5}$, both populations rescue so rarely that the absolute ΔP stays near zero across the sweep; resolving the flip cleanly at this mutation rate would require many more replicates than we have. Within the range where the model produces measurable rescue events, the qualitative shape of the flip is robust to μ: the mutation rate sets how often any rescue happens, but does not change which population gets the larger flow of core-born lineages arriving at the edge at high dose.

Across all three sweeps, the rescue flip is a robust qualitative feature of the model. The flip persists at every edge width l we tested (a sixteen-fold change in l), at every nonzero core activity b_c, and at every mutation rate μ for which the model produces measurable rescue events. The locations of the flip (the crossover doses) shift in directions that the race mechanism from Section 5 predicts: changes in l alter both the boundary speed and the duration of Phase 1, changes in b_c alter both how many core mutations are produced and how fast they drift to extinction, and

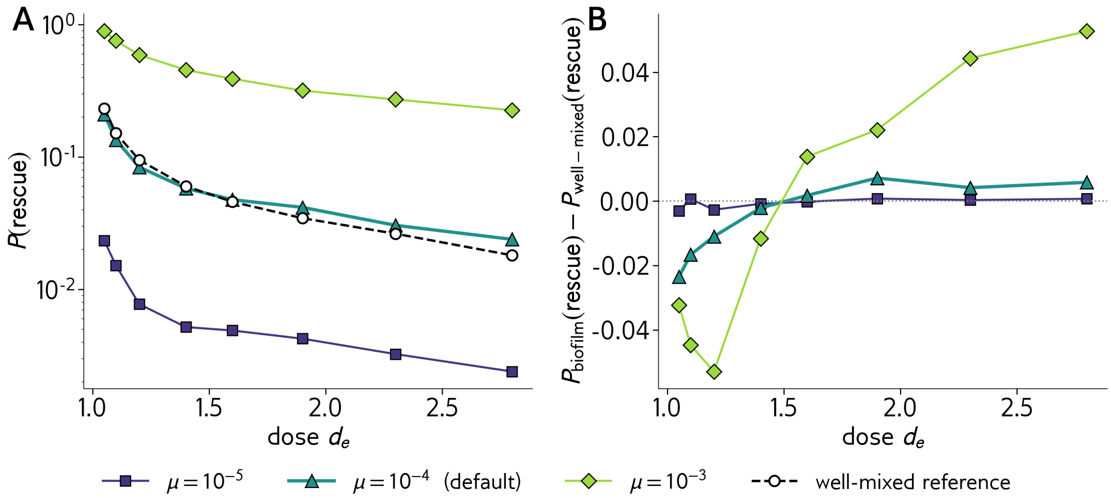

Figure 8: Sensitivity of the rescue flip to mutation rate μ. Panel A: $P(\text{rescue})$ vs dose for the biofilm at three values of $\mu \in \{10^{-5}, 10^{-4}, 10^{-3}\}$ and the well-mixed reference at $\mu = 10^{-4}$ (open circles, dashed). Panel B: difference relative to the well-mixed reference at each μ, $\Delta P(\text{rescue}) = P_{\text{BF}} - P_{\text{WM}}$, with the dotted horizontal line at zero. Each point averages $20{,}000$ independent simulations per parameter combination. *Parameters*: $N_0 = 1000$, $l = 100$, $b_c = 0.2$.

changes in μ scale the overall mutation supply without changing the race dynamics. The biofilm wins where these factors combine to produce a flow of core lineages arriving at the edge large enough to overcome the biofilm's per-mutation establishment penalty.

Sections 3 through 6 have now built the rescue flip, measured its mechanism, and tested its robustness across the three parameters that govern it. What remains is the question Section 1 actually opened with: what does this tell us about treating biofilm infections, and in particular about behavior-modifying interventions that disrupt biofilm spatial structure? The next section returns to that motivating question with the mechanism in hand.

7 Implications for biofilm treatment

Section 1 opened with a clinical question. Behavior-modifying interventions for biofilm infections, and in particular compounds that induce bacterial dispersal away from a biofilm into a more well-mixed, planktonic state, have been proposed as a way to reduce a bacterial population's capacity to evolve resistance under antibiotic pressure. The proposal does not require the intervention to kill cells directly; the hope is that disrupting biofilm spatial structure alone is enough to disarm the population, so that the surviving cells become more vulnerable to standard antibiotic treatment and to host immunity. The intuition behind that hope is that biofilm spatial structure helps the population evolve resistance, so disrupting that structure should reduce resistance evolution. Our findings from Sections 3 through 6 let us evaluate that intuition directly.

The answer depends on the antibiotic dose regime and on whether the biofilm's interior is metabolically active. At doses just above the minimum inhibitory concentration but below the crossover dose (our model places this near $d_e \approx 1.4$ at default parameters), the biofilm with an active core rescues less often than a well-mixed population of the same size (Section 3, Figure 3).

Dispersing the biofilm at this dose moves the bacterial population from a configuration in which resistance evolution is suppressed by spatial structure to one in which it is more efficient. The intervention is counterproductive: the patient would face a population more likely to evolve resistance after the intervention than the one they had before.

At doses above the crossover ($d_e > 1.4$ in our model), the biofilm with an active core rescues more often than a well-mixed population. Dispersing the biofilm at this dose moves the population in the opposite direction, from the configuration that favors resistance evolution to the one that suppresses it. The intervention does what the clinical hope requires: it reduces the chance that the surviving bacterial population evolves resistance. If the biofilm's interior is metabolically dormant, however, a third pattern holds. A dormant biofilm rescues less often than a well-mixed population at every dose tested (Section 6). Dispersing a dormant biofilm always increases the chance of resistance evolution, regardless of dose. The clinical hope that disrupting biofilm structure reduces resistance never holds in this case.

Several open questions can be addressed within this modeling framework. Two were promised in earlier sections and not yet pursued, but the existing simulator has the data to address them. First, why does the dormant biofilm produce the same number of mutations as the well-mixed reference despite living substantially longer (Figure 4 Panel A)? The simulator records the full population trajectory for every run, and the answer almost certainly lies in the integrated cell-time of the wild-type population: a longer-lived population that is, on average, smaller can produce the same total mutation budget as a shorter-lived population that is, on average, larger. Working this out from the model's deterministic limit would convert the supply equality from an observed coincidence into a derived consequence. Second, why does an edge-born mutation in a biofilm establish less often than the same mutation in a well-mixed population (Figure 4 Panel B)? The lineage tracking already in place would let us isolate the contributions of the conveyor belt's continuous replacement of wild-type cells (which displaces nascent R lineages with refreshed competitors) and of the density-factor coupling between core size and edge births (which keeps the edge near saturation in a way the well-mixed reference does not). Neither analysis requires changes to the simulator.

A third question is addressable with a small modification to the simulator: instead of comparing a biofilm to a separate well-mixed population, we could simulate the dispersal intervention itself in-run, by running the biofilm until a chosen intervention time and then converting it to a well-mixed (or partially dispersed) population mid-simulation. The rescue probability of such a hybrid run would be a direct in-model prediction of what a dispersal-inducing intervention does over the course of a treatment, including the dynamics of the transition itself.

A fourth set of questions requires extending the model. The 1D chain with a sharp step-function antibiotic profile is a stylization. Natural next versions include a 3D geometry with a graded drug-penetration profile, an explicit host immune response that finishes off the surviving wild-type population (Section 1's clinical hope quietly assumed this would happen), a heterogeneous core in which b_c varies with distance from the surface, and stochastic phenotypic switching between active and dormant states. Each adds complexity but stays within the same Gillespie framework, and each would let us test which assumptions the qualitative rescue flip actually depends on.

The model also makes two assumptions about the bacterial genotypes that, if relaxed, could change the qualitative results. First, every run starts with N_0 wild-type cells and zero resistant cells; new R lineages arise only through mutation during the simulation itself. Second, R cells have the same birth and death rates as wild-type cells in the core, so resistance carries no fitness cost in the absence of antibiotic. Real bacterial populations almost always carry low-frequency R variants as standing genetic variation before antibiotic exposure, and resistance often does carry a fitness cost when the antibiotic is absent: in a homogeneous well-mixed population, selection drives the standing R frequency downward over time. Recent work in spatial population genetics (Fruet and colleagues, Bitbol and colleagues, and related) has shown that spatial structure can act as a refuge that preserves these deleterious variants at higher frequencies than a well-mixed population would, by isolating them from competition with the fitter wild type. In our framework, this storage effect would be strongest at low b_c, where a largely dormant interior allows R lineages to persist without being actively selected against. If we allowed standing variation and a fitness cost for R outside the edge, we would expect the dormant biofilm to rescue more often than the well-mixed reference at low b_c, precisely the regime where our current model says the dormant biofilm has no rescue advantage at any dose. The dormant-loses-everywhere result is therefore likely specific to our no-standing-variation, no-fitness-cost assumptions; a more realistic treatment of pre-treatment dynamics would probably reverse the ordering of dormant biofilm and well-mixed at low b_c, with storage effects becoming the dominant axis promoting rescue rather than the mutation-supply axis we have studied here.

A separate biological assumption concerns the antibiotic's mode of action. The model treats the drug as purely bactericidal: it raises the death rate of wild-type cells in the edge (d_e becomes the swept variable) but leaves their birth rate b_e unchanged. Many real antibiotics are bacteriostatic (they slow or halt growth without killing cells directly), and many work via a mix of both modes. Under a purely bacteriostatic drug, wild-type cells in the edge would stop dividing rather than dying, so the edge would shrink only slowly through deaths at the baseline rate, the boundary would retreat much more slowly, and the supply of new edge-born mutations would drop because fewer edge births occur. The race from Section 5 between boundary retreat and core drift would still play out, but the boundary speed would no longer scale with dose in the way our bactericidal model assumes, and the dose-dependent shifts in the crossover could change in both magnitude and direction. Whether the qualitative rescue flip survives in a bacteriostatic regime is an open question the current simulator could answer with a small parameter change: making b_e dose-sensitive instead of d_e.

The clinical question from Section 1 does not have a single answer. The biofilm's spatial structure helps the population evolve resistance only at high antibiotic dose and with a metabolically active interior; in every other regime we tested, it hurts. A dispersal-inducing intervention is therefore context-dependent: useful in some clinical conditions, counterproductive in others, with the race mechanism in Section 5 determining which. The broader takeaway is that the double-edged sword from Section 3 captures a mechanistically grounded contingency: spatial structure in an evolving bacterial population under drug pressure is neither uniformly helpful nor uniformly harmful, and the direction of its effect on resistance evolution depends on a tractable race between mutation supply in the protected interior and mutation transport to the drug-exposed surface.

8 To-Do

The most pressing extensions to the model can be ranked by which would most directly test or change the qualitative result. Three stand out, and each is straightforward to implement on top of the existing simulator.

The first is standing variation with a fitness cost for R outside the edge. This is the extension most likely to flip the dormant-loses-everywhere result. The simulator currently starts every run with N_0 wild-type cells and no resistant cells, and treats R as neutral in the core. The minimal change is two parameters: a pre-treatment phase of duration T_{pre} during which the antibiotic is absent ($b_e = d_e = 1$ everywhere) and a fitness cost $c_R \in (0, 1)$ that reduces R cells' birth rate to $b \cdot (1 - c_R)$ throughout the pre-treatment phase, where b is the compartment-appropriate birth rate. During T_{pre}, mutations arise at rate μ per wild-type birth, R cells incur the fitness cost wherever they are alive, and a standing R frequency builds up that depends on μ, c_R, and the spatial structure. At $t = T_{\mathrm{pre}}$ the antibiotic switches on, current dynamics resume, and rescue is measured as before. The expected output: in the well-mixed population, standing R is selected down to a low frequency of order μ/c_R. In the dormant biofilm, R mutations produced in the edge can be pushed into the core via the conveyor belt, where there is no cell turnover and therefore no selection against them; over time, the dormant core preserves a reservoir of R cells that would have been culled in a well-mixed population. When treatment starts, the dormant biofilm enters with this preserved reservoir, and may rescue more often than the well-mixed reference at low dose, where our current model says it always loses. The natural sweep is c_R across a small range with T_{pre} long enough for the standing R frequency to equilibrate.

The second is the antibiotic's mode of action. The simulator's d_e is currently the only dose-sensitive parameter; b_e stays at 1. The minimal change is to make both edge rates dose-sensitive, with the choice of which actually varies selecting the drug's mode. In a purely bactericidal regime (current), the dose maps to d_e alone and b_e stays at 1. In a purely bacteriostatic regime, the dose maps to b_e instead, with b_e shrinking from 1 toward zero as dose increases, and d_e remaining at 1. The boundary speed is $s_e \cdot l$ with $s_e = d_e - b_e$ in either case. In the bactericidal case s_e grows linearly with dose; in the bacteriostatic case s_e saturates near 1 as $b_e \to 0$, so the boundary speed has a hard ceiling no matter how high the dose is pushed. The race from Section 5 would therefore have a fundamentally different high-dose limit: the boundary cannot run away from drift as quickly under a bacteriostatic drug as it can under a bactericidal one. The natural experiment is to run the same dose range under each mode and compare crossover doses, advantage magnitudes, and whether the dose-dependent flip survives in shape or is replaced by a saturating profile at high dose.

The third is dispersal as an in-simulation event. Currently the biofilm and well-mixed populations are simulated as separate runs. The minimal change is to allow a transition from biofilm to well-mixed at a specified time T_{disp} within a single run, by setting $l \to N(T_{\mathrm{disp}})$ at that moment so that every remaining cell is reclassified as edge. After T_{disp}, the simulation continues with no core. The rescue probability of such a hybrid run is a direct in-model prediction of what a dispersal intervention does over the course of a treatment, including the dynamics of the transition itself. The natural sweep is T_{disp} across the dose response: intervening very early (close to $t = 0$) should produce dynamics close to a pure well-mixed run from the start; intervening very late (after most rescue events have occurred) should produce dynamics close to a pure biofilm run. The interesting regime is the intermediate one, and whether there is a T_{disp} that minimizes rescue probability at a given dose, which would correspond to the optimal timing of a dispersal intervention.